References:

- https://www.educationworld.com/a_curr/strategy/strategy036.shtml
  https://www.teachhub.com/teaching-strategies/2016/10/the-jigsaw-method-teaching-strategy/

1/12/22

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- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students feel that they have improved self-learning.
    - They learn how to communicate with team members and work together.
    - They were completely involved
  - ***Benefit of the practice:***
    - Every student got equal opportunity to come forward to take part in this activity.
    - They felt easy in solving level 3 questions in Assignments and Internal Assessment test
    - The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correct
  - ***Challenges faced in implementation:***
    - Few students hesitated to come forward to present the results and conclusions of the disscussion
    - The main challenge faced is that few students not exposed to flipped class room.

References:

- https://www.teachthought.com/learning/a-flipped-classroom/
- Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.

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Department of Electronics and Communication Engineering
Academic Year 2022 – 2023 (Odd Semester)
Degree, Semester & Branch: V semester B.E. ECE B
Course Code & Title: EC8553 Discrete Time Signal Processing
Name of the Faculty member (s): P.Venkatesh

Innovative Practice Description

- **Unit / Topic:** IV / FINITE WORD LENGTH EFFECTS
- **Course Outcome:** CO4 : The students will be able to analyse the finite Word length effects in digital systems.
- **Topic Learning Outcome:** TLO 15: To Explain and classify the quantization errors
- **Activity Chosen:** Jigsaw
- **Justification:**
The Jigsaw Strategy is an efficient way to learn the course material in a cooperative learning style. The jigsaw process encourages listening, engagement, and empathy by giving each member of the group an essential part to play in the academic activity. Group members must work together as a team to accomplish a common goal each person depends on all the others.
- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**
The lessons are divided into subcategories. Students are divided into groups of four or five students. Then each small group would be created with one student receiving one Subcategory of the lesson. For this method, each small group gets the same set of Subcategories. Once individuals have researched their own subcategory, they will meet with individuals from the other small groups with the same topic to better develop their understanding and become experts of the subcategory. Each student would then return to their original group and teach their subcategory to the rest of their small group.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO8	PO9	PO10	PSO3
CO4	2	3	3	2	3	3	2

- **PO / PSO mapped:**

novative ractice	PO1	PO2	PO3	PO8	PO9	PO10	PSO1
	2	3	3	2	3	3	2
ification for relation	Finite word length effects requires the knowledge of engineering fundamentals such as	The course outcome is highly correlated, since identification of	The filter performance needs analysis of Quantization noise to solve	Identify tenets of the professional code of ethics while doing activity-	A Collaborative activity – Jigsaw helps the students to Present results as a	The topics deals with presentations of fixed- and floating-point representation. This task is	The concept finite word length effect such as quantitation noise, limit cycle

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V semester B.E. ECE B

Course Code & Title: EC8553 Discrete Time Signal Processing

Name of the Faculty member (s): P. Venkatesh

Innovative Practice Description

- **Unit / Topic:** Unit V / INTRODUCTION TO DIGITAL SIGNAL PROCESSORS
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 21
- **Activity Chosen:** Flipped Classroom
- **Justification:**
  It allows students to learn in their own pace, it encourages students to actively engage with lecture material, it frees up actual class time for more effective, creative and active learning activities and students take control and responsibility for their learning. Also the practice was chosen because the topic is simpler and interesting for the students to refer and learn by themselves

- **Time Allotted for the Activity:** 45 minutes
- **Details of the Implementation:**

ovative actice	PO1	PO2	PO5	PO8	PO9	PO10	PO12	PSO2	PSO3
	3	3	3	2	3	3	1	1	3
fication for elation	DSP Processors requires the knowledge of engineering fundamentals such as processor flow, Accumulators, storage devices. This course outcome is highly correlated.	Programming using DSP processors needs to Identify the engineering and other relevant knowledge that applies to a given problem. So, the course outcome is highly correlated	The course outcome is highly correlated since, the functionalities of DSP processors involve Identification of modern engineering tools such as code composer, fixed and floating-point processor kit.	Flipped Class activity involves presentation which apply moral & ethical principles to known case studies while collecting materials related to the topic. The course outcome is moderately correlated.	Flipped class activity needs to demonstrate effective communication, problem-solving, conflict resolution and leadership skills and listen to other members. The course outcome is highly correlated.	The topics application of DSP deals with presentations and preparation of contents which requires written skills. This task is accomplished through Flipped class activity therefore course outcome is correlated highly	The concept of processors is important to keep current trends regarding new developments in the field of information technology. This course outcome is correlated with low.	The course outcome is moderately corrected because the concept of FFT using DSP processor is required in design of ARM processors.	The knowledge of DSP processors and its architecture is useful in designing VLSI and Communicatio systems. The course outcom is highly correlated.

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Department of Electronics and Communication Engineering

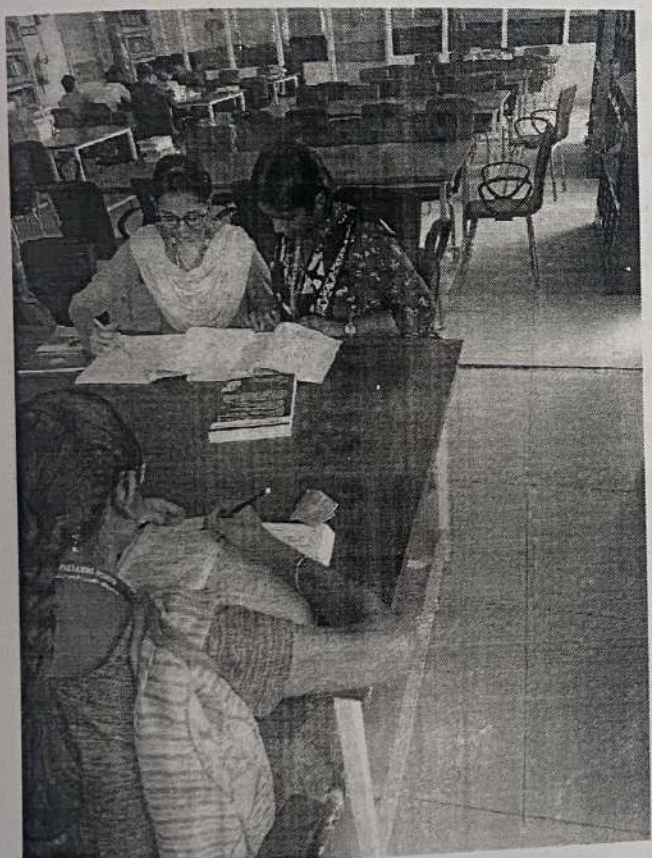
Academic Year: 2022- 2023 (Odd Semester)

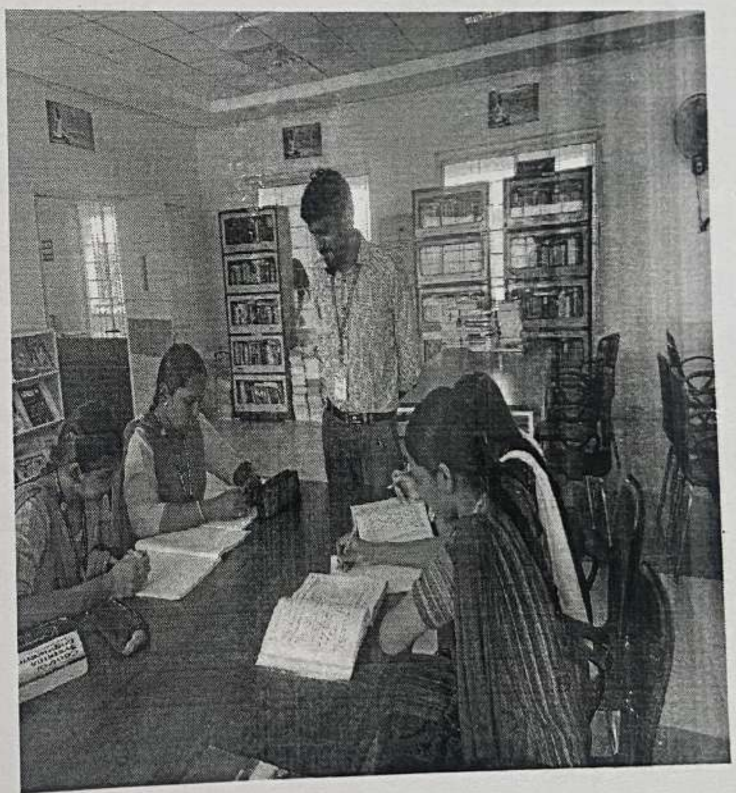
Degree, Semester & Branch : **V Semester B.E. ECE B**

Course Code & Title : **EC8553 Discrete Time Signal Processing**

Name of the topic : **Frequency sampling techniques**

Name of the activity : **Open book test (Library)**





31/10/22

HOD Incharge

31/10/22

Signature of the Faculty member

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Department of Electronics and Communication Engineering

Academic Year: 2022- 2023 (Odd Semester)

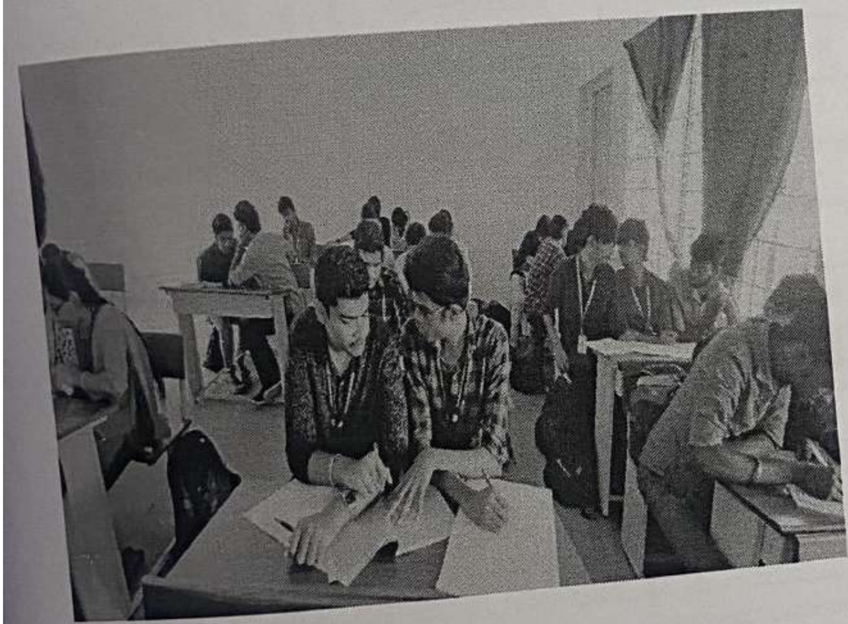
Degree, Semester & Branch : V Semester B.E. ECE B

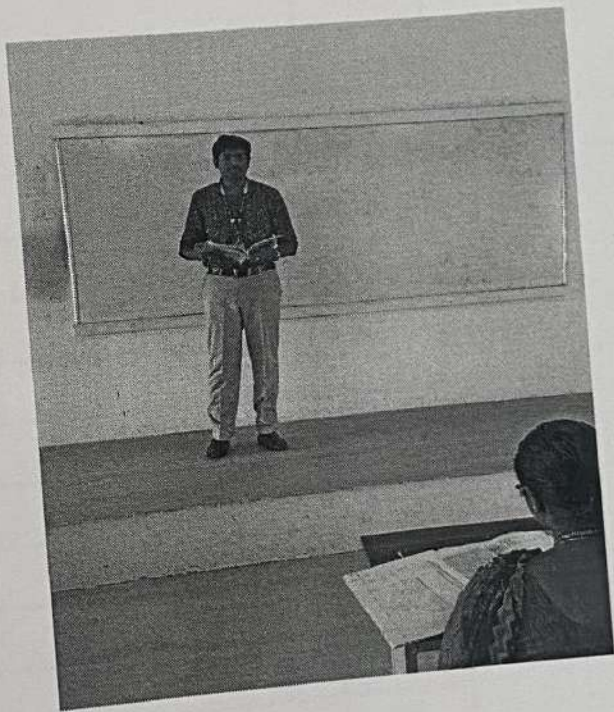
Course Code & Title : EC8335 Discrete Time Signal Processing

Name of the Faculty member : P.Venkatesh

Name of the topic : Characteristics of commonly used analog filters

Name of the event : Think Pair Share





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RAMCO INSTITUTE OF TECHNOLOGY

Department of Electronics and Communication Engineering

Academic Year: 2022- 2023 (Odd Semester)

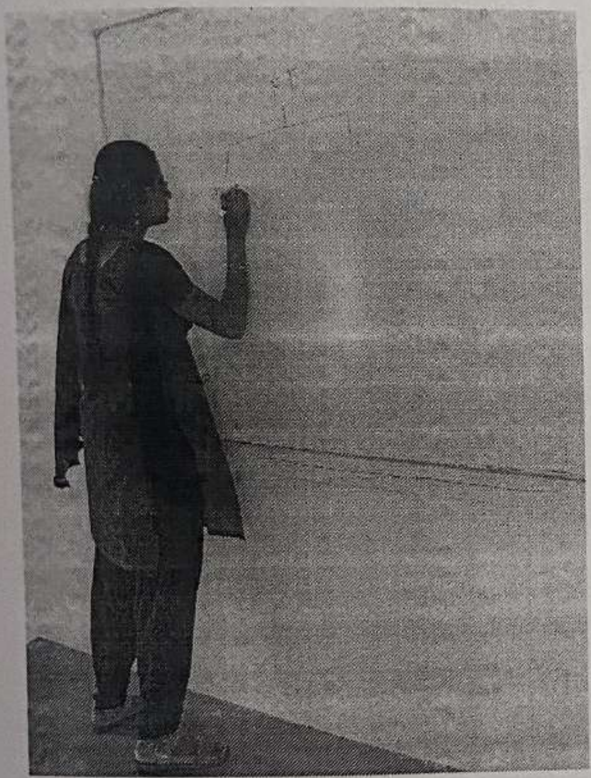
Degree, Semester & Branch : V Semester B.E. ECE B

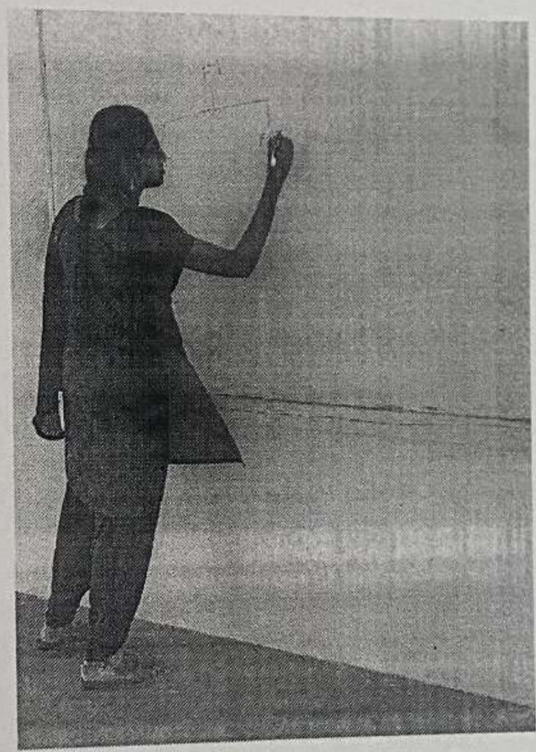
Course Code & Title : EC8335 Discrete Time Signal Processing

Name of the Faculty member : P.Venkatesh

Name of the topic : Linear Filtering using FFT & Revision

Name of the event : Concept Mapping





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